Recall

Work from memory, then check with your tutor.

R1. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

R2. Find $\frac{3}{4} - \frac{1}{3}$.

Answer: _____

R3. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

Answer: _____

R4. Find $3 \times \frac{2}{3}$.

Answer: _____

S. Return here near the end of the lesson

To compare two mixtures fairly, I can make ___ the same.

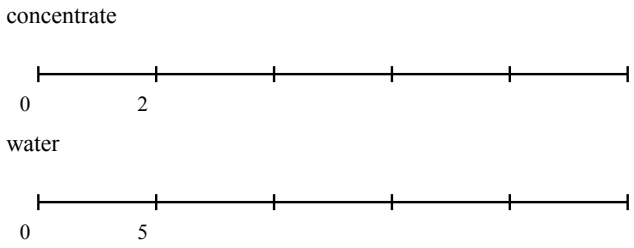
Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Concentrate : water is 2 : 5. Complete pairs (2, 5), (4, __), and (6, 15).

concentrate	water
2	5
4	?
6	15

P2. For that 2 : 5 mixture, how much water matches 10 cups concentrate? Solve using a table and a double number line.



P3. Mix A uses 3 cups syrup to 4 cups water. Mix B uses 5 cups syrup to 8 cups water. Which has more syrup for the same water?

Answer: _____

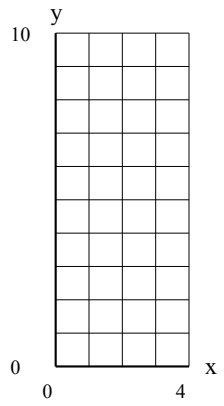
Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Yellow : blue paint is 2 : 3. There are 35 cups in all. How much of each color?

Answer: _____

P5. Plot (1, 3), (2, 6), and (3, 9). Let x be bags and y be marbles. Does their straight line also pass through (0, 0)?



x: bags; y: marbles

F. Green : white is 4 : 1. There are 30 beads. Five equal parts means __ beads per part. Green = __; white = __.

Answer: _____

Ratio tables and comparisons

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

Answer: _____

Mixed review

M1. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

Answer: _____

M2. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

Answer: _____

M3. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Mix A has 4 cups juice and 3 cups water. Mix B has 6 cups juice and 5 cups water. Which has the larger fraction of juice?

Answer: _____

Ratios and matching amounts

The big idea

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

Words to know

Ratio: A comparison of two amounts in a stated order. Example: 4 red to 3 blue is 4 : 3.

Equivalent ratios: Ratios that describe the same relationship. Example: 4 : 3 and 8 : 6 are equivalent.

Scale factor: The number used to multiply both amounts. Example: From 4 : 3 to 12 : 9, the factor is 3.

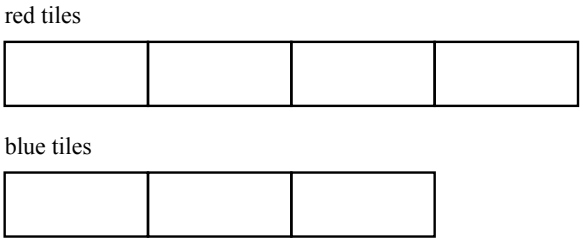
Ratio table: A table of matching amounts with the same ratio. Example: (2, 5), (4, 10), and (6, 15).

Double number line: Two aligned scales showing matching amounts. Example: Two cups concentrate matches five cups water.

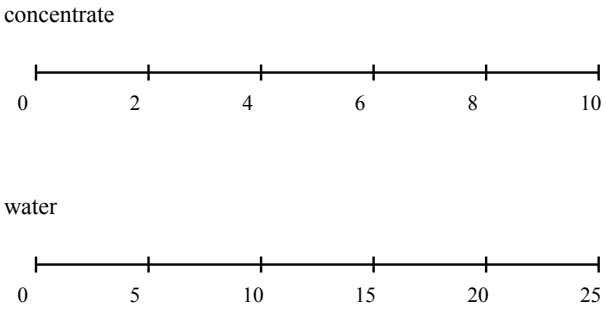
Part-to-whole: A comparison between one part and the total. Example: 2 red and 3 blue makes red $\frac{2}{5}$ of all tiles.

How it works

Repeat the whole 4 : 3 group. Equivalent ratios multiply or divide both quantities by the same factor.



Two cups concentrate pairs with five cups water. A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.



Worked examples

1. A mix uses 3 cups oats for every 2 cups seeds. Find seeds for 12 cups oats.

12 ÷ 3 = 4 batches.

Each batch has 2 cups seeds.

4 × 2 = 8 cups seeds.

Answer: 8 cups seeds

2. Orange : purple tiles is 3 : 2. There are 25 tiles. Find each count.

3 + 2 = 5 equal parts.

25 ÷ 5 = 5 tiles per part.

Orange: 3 × 5 = 15. Purple: 2 × 5 = 10.

Answer: 15 orange, 10 purple

Common mistakes

Adds the same count to both colors. Why: Confuses additive and multiplicative change. Fix: Build identical batches.

Says 4 : 3 means four out of three. Why: Confuses part-to-part and part-to-whole. Fix: Count the total seven tiles before naming the red fraction.

Compares mixtures by subtraction. Why: A difference ignores mixture size. Fix: Use the same amount of water in both mixtures.

Try it

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

2. Is 6 : 10 equivalent to 9 : 15?

3. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

4. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

5. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

Answers

1. 5 : 2; 2 : 5 2. Yes 3. 21 4. 32 laps 5. Yes

Where this connects

This builds on equal groups and fraction comparison. Next, unit rates and percents describe one-unit and hundred-unit comparisons.